Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

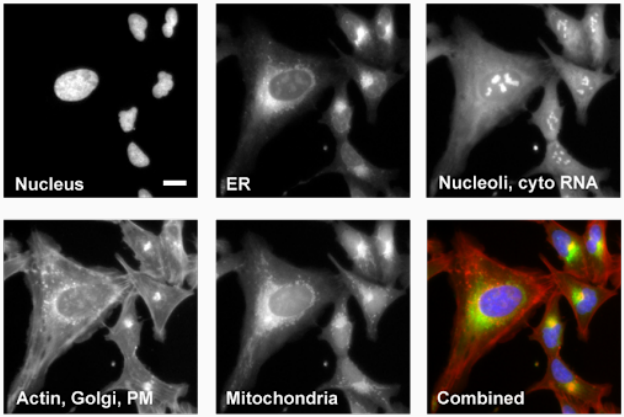
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University

10 November 2026

What is Cell Painting?

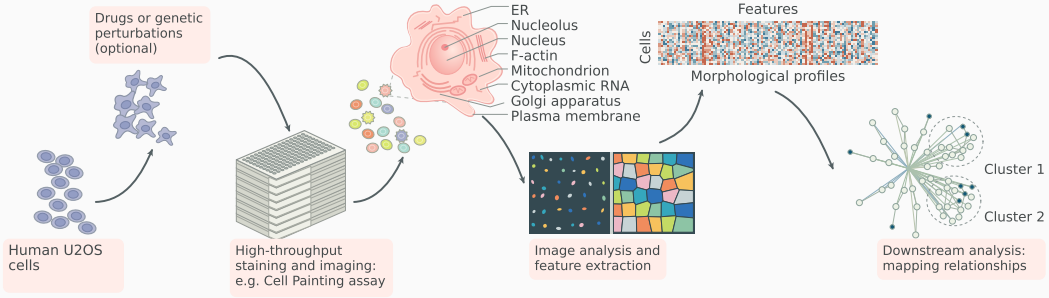
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

Two genes, two routes to mitochondrial shape

OPA1: mitochondrial fusion

- Helps mitochondria fuse.
- Loss is expected to produce shorter, fragmented networks.

DNM1L / DRP1: mitochondrial division

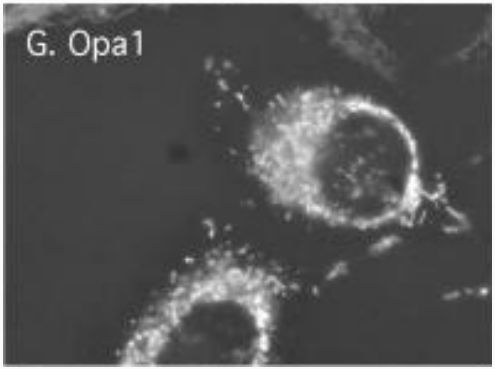
- Helps mitochondria divide.
- Loss is expected to produce longer, more-connected networks.

Inspect each knockout beside its plate-matched negative control.

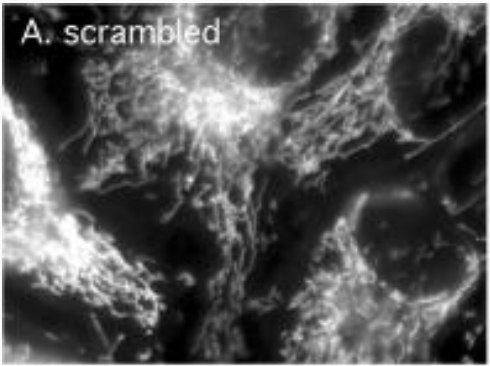
Literature expectations: Griparic et al., JCB (2007), Fig. 3; Smirnova et al., MBoC (2001).

Published example: OPA1 depletion fragments mitochondria

OPA1 siRNA



Scrambled siRNA

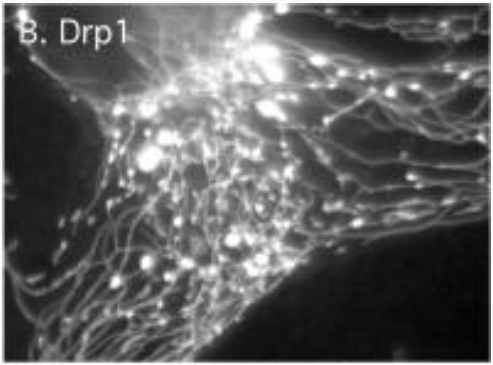


HeLa cells; MitoTracker. OPA1 depletion for 2 days.

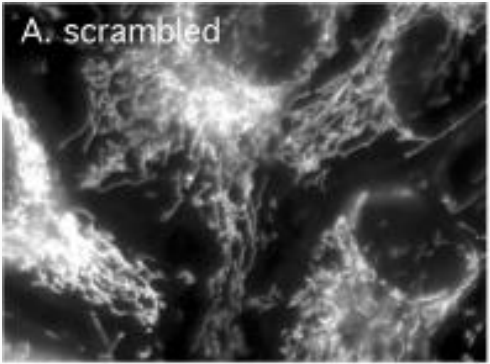
Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3G/A. Published microscopy, cropped; CC BY-NC-SA 4.0.

Published example: DRP1 depletion increases connectivity

DNM1L / DRP1 siRNA



Scrambled siRNA

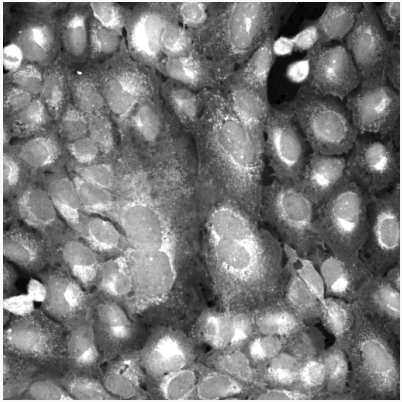


HeLa cells; MitoTracker. DRP1 depletion for 2 days.

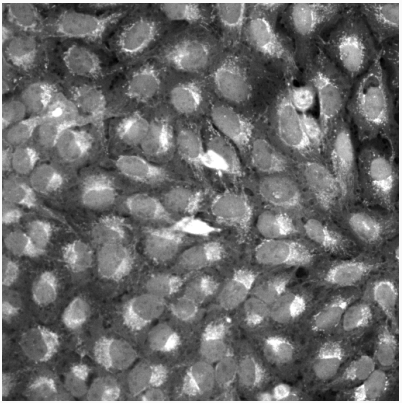
Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3B/A. Published microscopy, cropped; CC BY-NC-SA 4.0.

JUMP example: OPA1 knockout

OPA1 CRISPR knockout



No-guide negative control

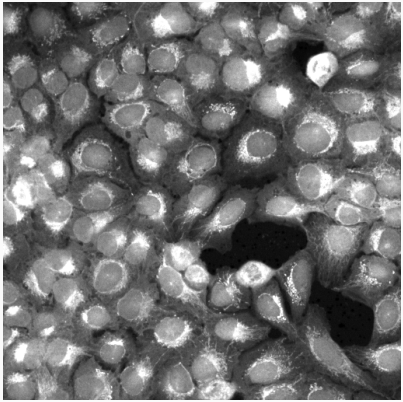


JUMP Mito channel; source_13, CP-CC9-R1-06, D20 / B02, field index 0.

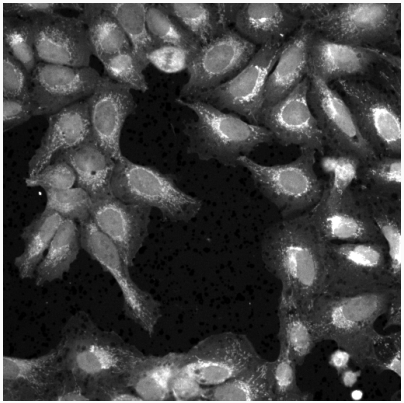
Central 512 × 512-pixel crops; both panels share intensity limits 219–3985. JUMP data.

JUMP example: DNM1L knockout

DNM1L CRISPR knockout



No-guide negative control



JUMP Mito channel; source_13, CP-CC9-R1-02, P18 / B02, field index 0.

Central 512 × 512-pixel crops; both panels share intensity limits 187–4463. JUMP data.

CellProfiler detects mitochondrial feature changes

Gene	Feature	Cohen's d
OPA1	Mito texture correlation	$+3.56$
DNM1L	Mito-skeleton branch ends	-5.95
DNM1L	Mito granularity	$+4.09$
DNM1L	Mito-skeleton total length	-3.55

The image differences accompany changes in mitochondrial organization.

Workflow 1: follow the genes from papers to JUMP

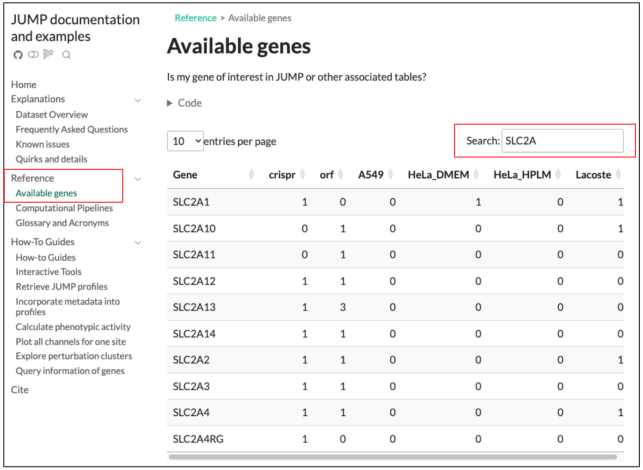
Search: OPA1 and DNM1L

Gene	CRISPR knockout	ORF overexpression
OPA1	Available	Available
DNM1L	Available	Available

- Follow each gene to its exact perturbation identifier.
- Open the feature, match and image tables.
- Start the image comparison with CRISPR and its genetic controls.

JUMP Hub gene availability; coverage reference

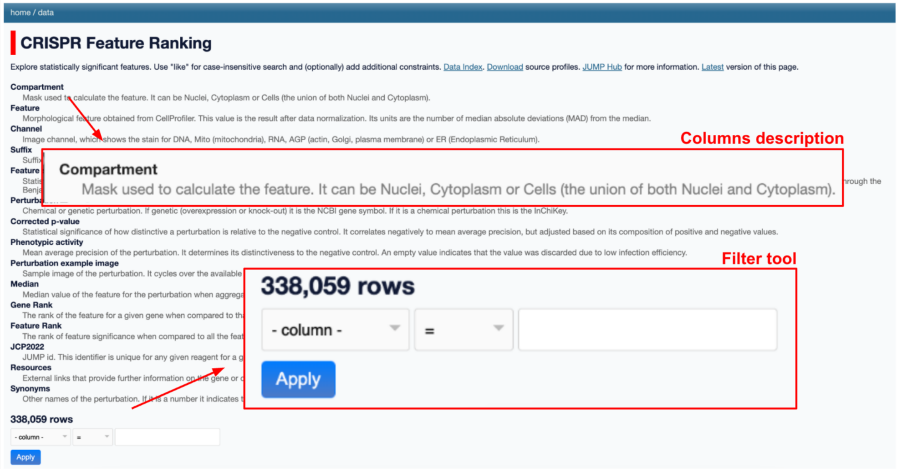
Gene availability: search a gene family



Gene	crispr	orf	A549	HeLa_DMEM	HeLa_HPLM	Lacoste
SLC2A1	1	0	0	1	0	1
SLC2A10	0	1	0	0	0	1
SLC2A11	0	1	0	0	0	0
SLC2A12	1	1	0	0	0	0
SLC2A13	1	3	0	0	0	0
SLC2A14	1	1	0	0	0	0
SLC2A2	1	1	0	0	0	1
SLC2A3	1	1	0	0	0	0
SLC2A4	1	1	0	0	0	1
SLC2A4RG	1	0	0	0	0	0

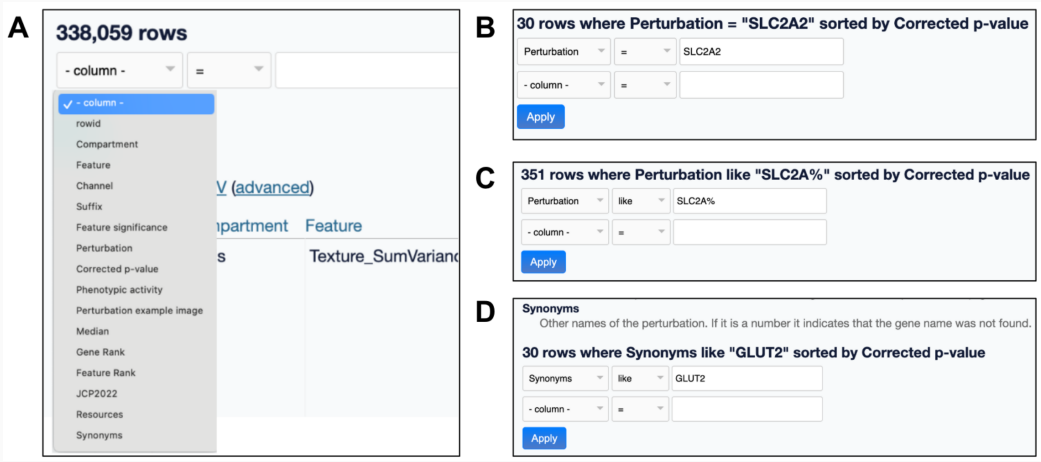
Read the dataset columns: find the available CRISPR and ORF perturbations.

Read the column definitions first



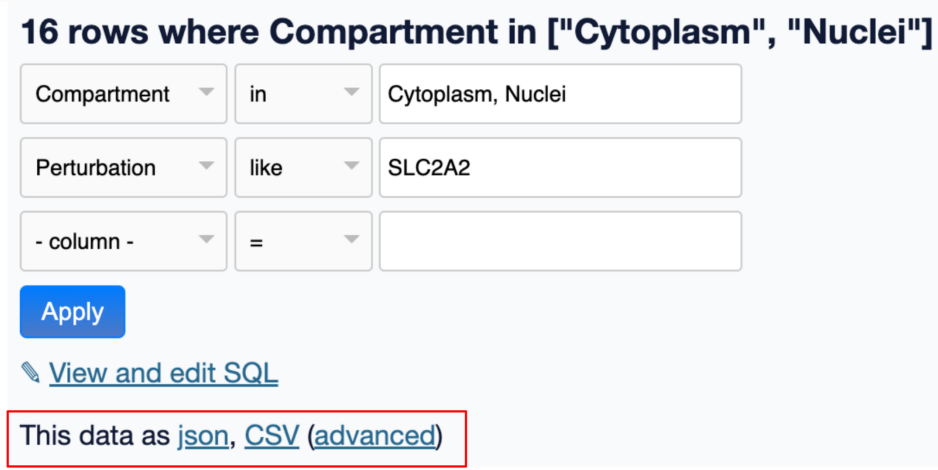
Use the red boxes: read the field definitions, then choose the filter for your question.

Filtering: exact gene, family and synonym



Compare the panels: select SLC2A2, expand to the SLC2A family, or search the GLUT2 synonym.

Keep the relevant compartments and save the view



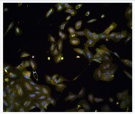
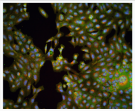
Save the query context: selected gene, compartments, identifiers and source link.

Workflow 2: read the returned phenotype

1. Read overall activity and the changing CellProfiler features.
2. Compare CRISPR and overexpression without assuming opposite effects.
3. Follow similar profiles to their annotations and images.
4. Bring in genetic controls from the same source and plates.

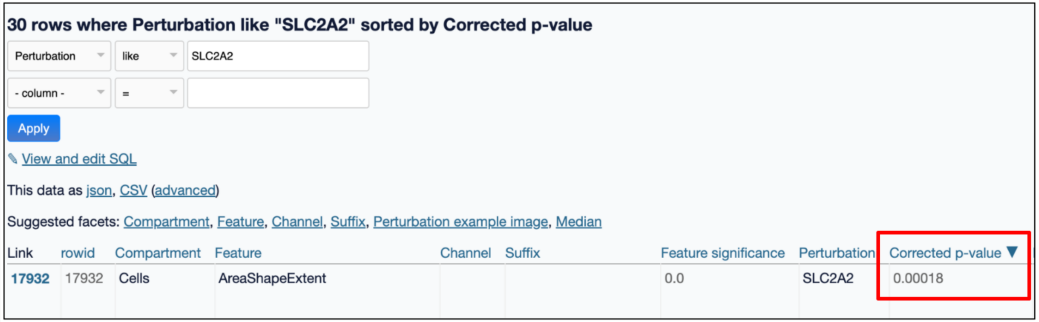
Tutorial navigation examples: SLC2A2 for tables; MYT1 for the gallery.

Understand a feature-table result

Compartment	Feature	Channel ▲	Suffix	Feature significance	Perturbation	Corrected p-value	Phenotypic activity	Perturbation example image	Median	Gene Rank	Feature Rank
Cells	Intensity_MassDisplacement	RNA		1e-05	SNRNP200	0.00018	1.0		-0.965	999999	27
Nuclei	Texture_AngularSecondMoment	RNA	_5_03_256	0.0	CECR1	0.00018	0.77241		-0.203	999999	22

Read across the row: perturbation, feature, compartment, channel and statistical summary.

Overall activity: SLC2A2 CRISPR



30 rows where Perturbation like "SLC2A2" sorted by Corrected p-value

Perturbation | like | SLC2A2

- column - | = |

Apply

View and edit SQL

This data as json, CSV (advanced)

Suggested facets: Compartment, Feature, Channel, Suffix, Perturbation example image, Median

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation	Corrected p-value ▼
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2	0.00018

Read the boxed corrected p-value: it summarizes overall perturbation activity.

Describe the changes: SLC2A2 CRISPR

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2
21804	21804	Nuclei	NeighborsSecondClosestDistance		_1	0.0	SLC2A2
22777	22777	Cells	NeighborsPercentTouching		_Adjacent	0.0	SLC2A2
33343	33343	Cytoplasm	RadialDistributionFracAtD		_mito_tubeness_12of20	0.0	SLC2A2

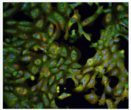
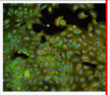
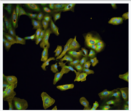
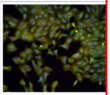
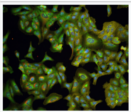
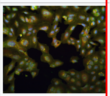
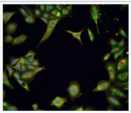
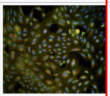
Read feature significance, rank and effect size to describe the morphological changes.

Compare perturbation types: SLC2A2 ORF

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance ▼
394755	394755	Cytoplasm	AreaShapeMinorAxisLength			8e-05
293369	293369	Cells	AreaShapeMinorAxisLength			0.00011
126201	126201	Cells	AreaShapeMinFeretDiameter			0.00036
292792	292792	Cytoplasm	AreaShapeMinFeretDiameter			0.00036

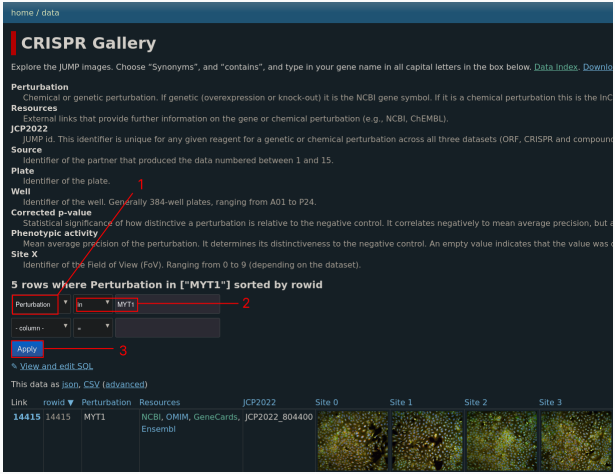
Compare the two levels: overall perturbation activity and the individual feature results.

Explore related profiles: SLC2A2 CRISPR

Link	rowid	Perturbation	Match	Perturbation example image	Match example image	Perturbation-Match Similarity ▲	Match resources
210190	210190	SLC2A2	RAD51			0.621	NCBI, OMIM, GeneCards, Ensembl
316101	316101	SLC2A2	BACE1			0.614	NCBI, OMIM, GeneCards, Ensembl
163302	163302	SLC2A2	FANCM			0.61	NCBI, OMIM, GeneCards, Ensembl
182659	182659	SLC2A2	TONSL			0.61	NCBI, OMIM, GeneCards, Ensembl

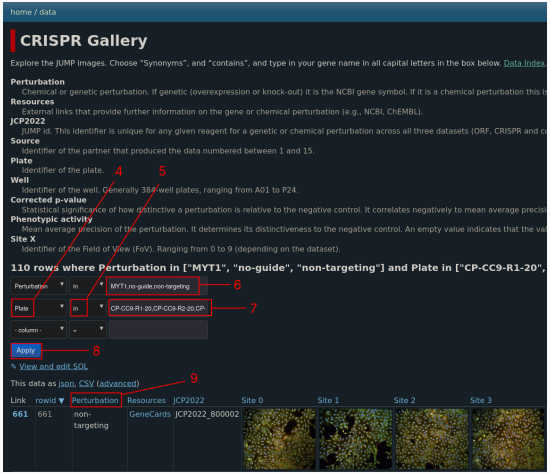
Sort by signed similarity, then follow the annotation links and inspect the example images.

Find the perturbed images: MYT1



Follow steps 1–3: select the perturbation, apply the filter, and identify its source and plates.

Bring in matched controls: MYT1



Follow steps 4–9: select the same plates and compare cell appearance, density, staining and focus.

Bring the workflow together with an agent

Start with a biological question

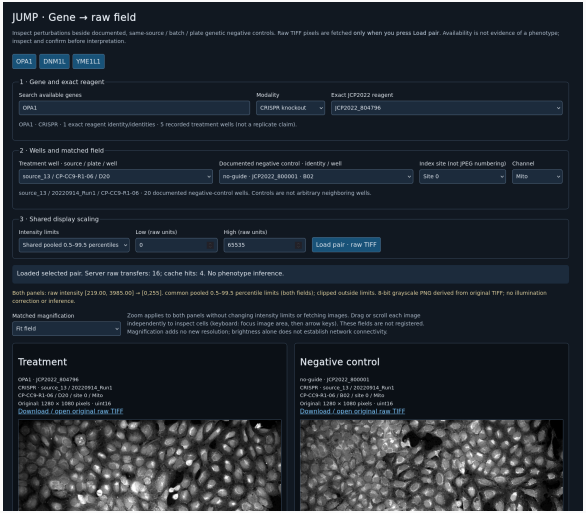
How does disrupting mitochondrial fusion or division change cell images? Find primary studies for OPA1 and DNM1L and explain the expected phenotypes.

Build a tool to inspect the data:

“Let me choose a gene and CRISPR reagent, load its raw images beside a plate-matched negative control, and zoom into cells using shared intensity limits.”

Output: cited expectations, traceable image pairs and an interactive viewer.

AI-assisted tools: gene and raw-image explorer



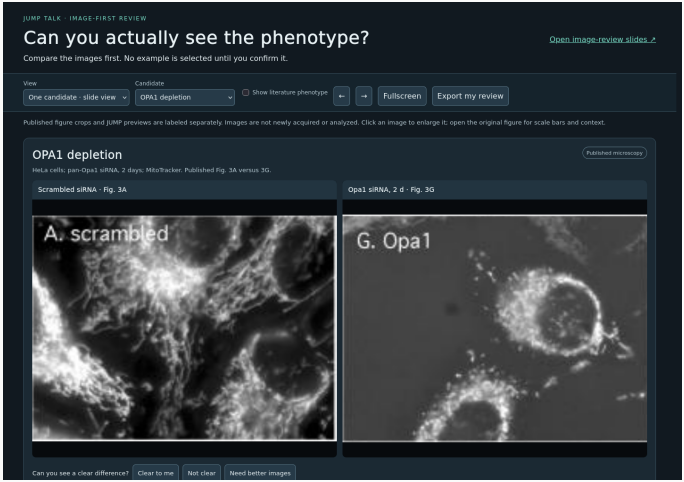
Explorer images: JUMP.

From a gene to its images

- Select a gene and perturbation.
- Load raw images with plate-matched controls.
- Zoom into cells with shared intensity limits.

You define the question. The agent builds. You inspect and refine.

AI-assisted tools: phenotype comparison viewer



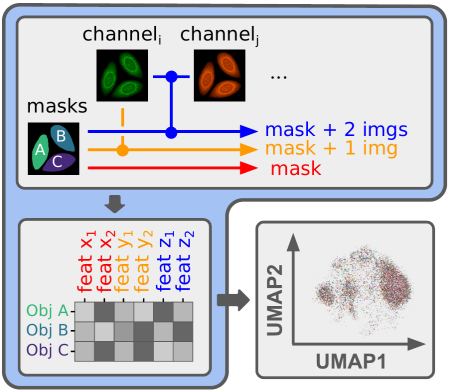
Microscopy: Griparic et al., JCB (2007), Fig. 3A/G, cropped; CC BY-NC-SA 4.0.

Compare before choosing

- Compare candidate image pairs.
- Inspect original figures and sources.
- Record which differences are visible.

Published figures and JUMP previews remain clearly labeled.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

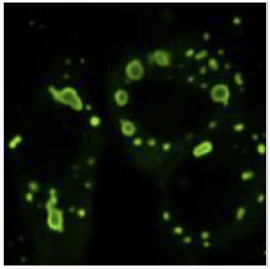
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

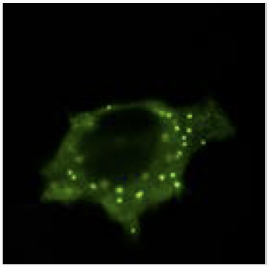
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

BROAD INSTITUTE

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Anne E. Carpenter
Shantanu Singh

Cimini Lab
Beth Cimini
Paula Llanos

PURDUE UNIVERSITY

Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners



JUMP supporting partners



JUMP collaborators

Scientific:
- Horizon Discovery, a PerkinElmer Company

Administrative:
- Pistoia Alliance

Academic:
- Umeå University
- Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

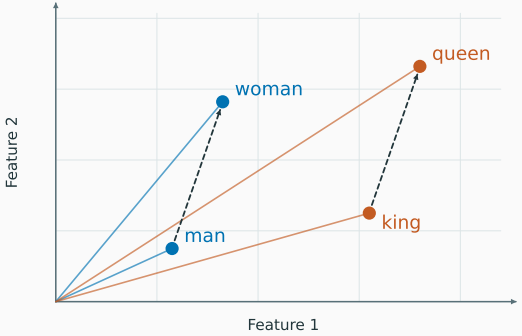
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

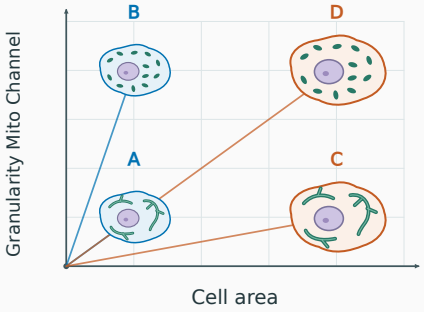
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

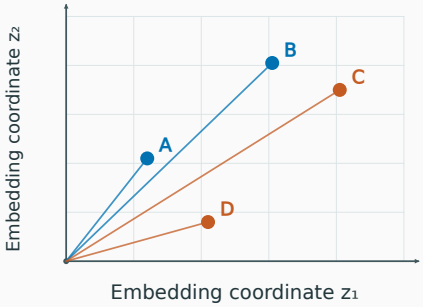


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



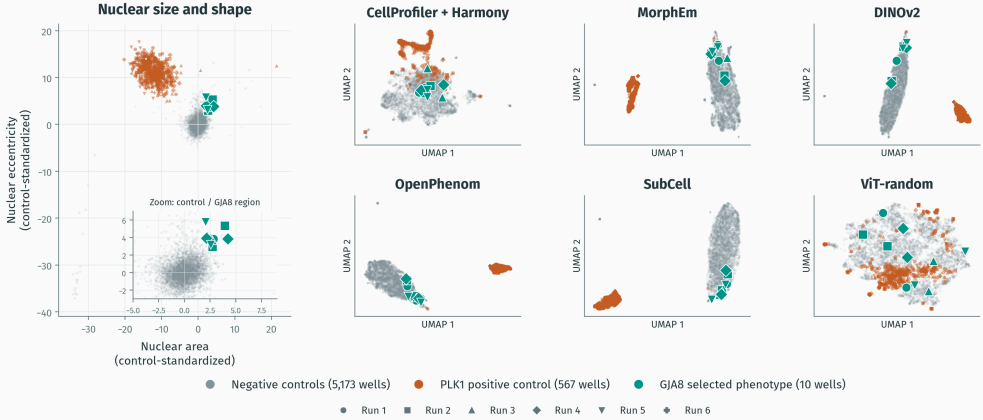
Control

Perturbation

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

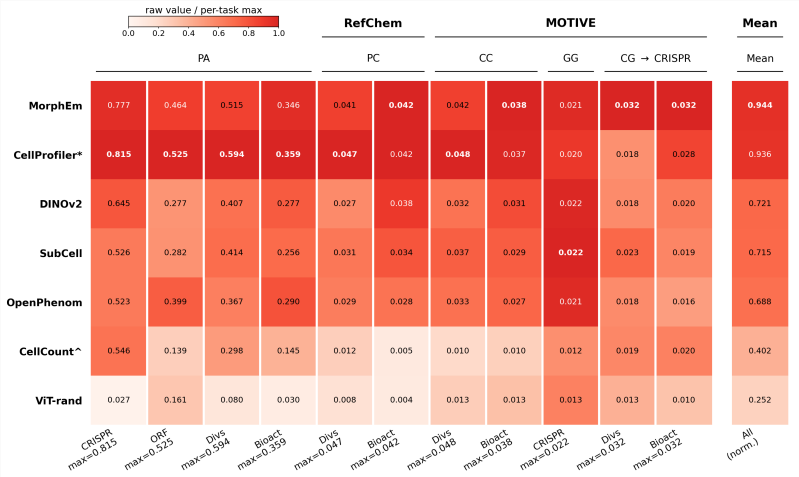
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

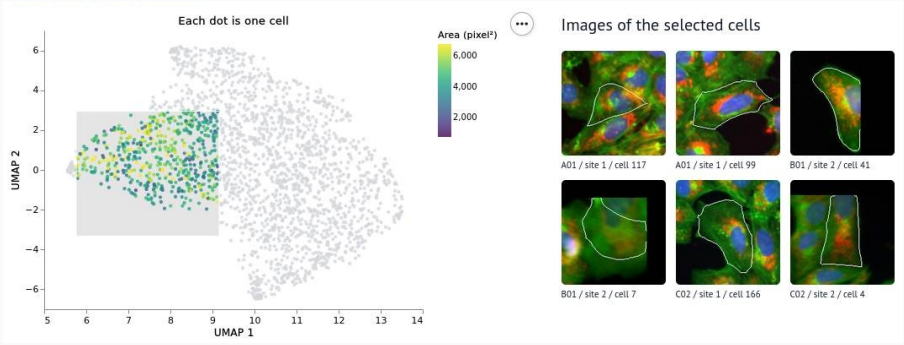


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

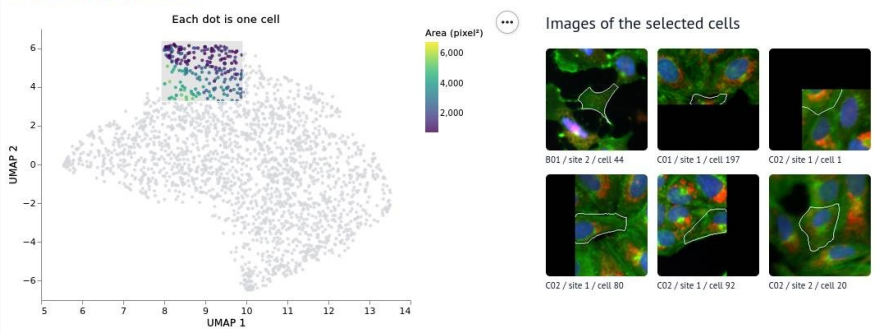
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the compound queries; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording